

Unit 10

Model Choice and Diagnostics

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Residual plots, Q–Q plots, model assumptions, transformations, outliers, overdispersion, and sensitivity analysis.
Central habit	Do not ask only whether a test is significant. Ask whether the model is a reasonable description of the data-generating process.

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1 Purpose of Unit 10

Earlier units introduced descriptive statistics, distributions, hypothesis tests, comparisons, ANOVA, chi-square tests, and regression. Unit 10 asks a deeper question:

Was the chosen model appropriate enough for the conclusion?

This is where many student investigations become weak. Students run a test, copy the p-value, and stop. But every statistical result is conditional on assumptions about the data and the model.

Core thesis

A statistical method is not only a calculation. It is a model of how the data were generated. Diagnostics ask whether the data strongly contradict that model.

By the end of this unit, students should be able to:

1. explain why model choice is a scientific judgement, not a software menu choice;
2. use residual plots to check linearity and equal spread;
3. explain what Q-Q plots check and what they do not prove;
4. distinguish outliers, high-leverage points, and influential points;
5. justify transformations such as log, square-root, or reciprocal transformations;
6. recognize overdispersion in biological count data;
7. compare plausible models without pretending that one model is final truth;
8. write conclusions that are conditional on model assumptions.

2 The model is part of the claim

A model is a simplified description of the data-generating process. For example:

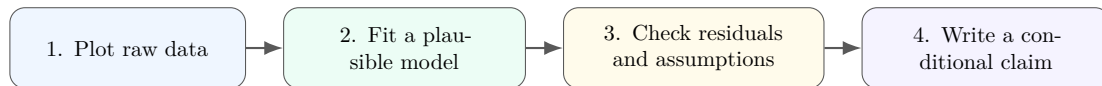
Situation	Possible model	Implied assumption
Calibration curve	Linear regression	response changes approximately linearly with concentration in the tested range
Titration volumes	Normal error model	repeated endpoint errors are roughly symmetric around a mean
Survival counts	Binomial model	each organism has a yes/no outcome with a similar probability within group
Colony counts	Poisson model	colonies arise as independent events with mean close to variance
Skewed concentrations	Log-normal or transformed model	multiplicative variation is more reasonable than additive variation

Bad habit

Do not write: “I used a t-test because the software has a t-test button.” A defensible analysis says why the model is reasonable for the variable, the design, the sample size, and the graph.

3 A diagnostic workflow

A practical diagnostic workflow has four stages.



Reading the graph: blue dots are observed values, the dark line is the fitted line, and the red vertical segments show residuals.

Scientific discipline

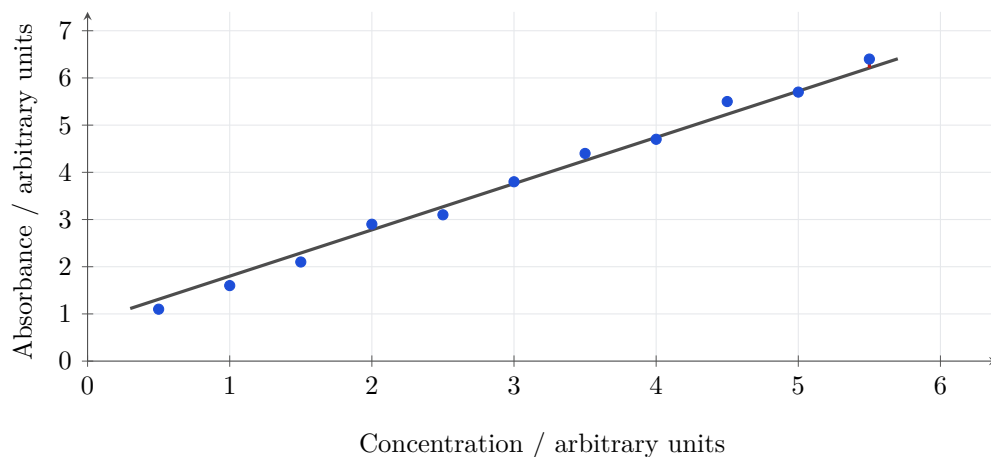
The point of diagnostics is not to achieve perfect assumptions. The point is to ask whether the model is good enough for the intended claim.

4 Residuals: what the model failed to explain

For a regression model, a residual is:

$$\text{residual} = \text{observed value} - \text{predicted value}.$$

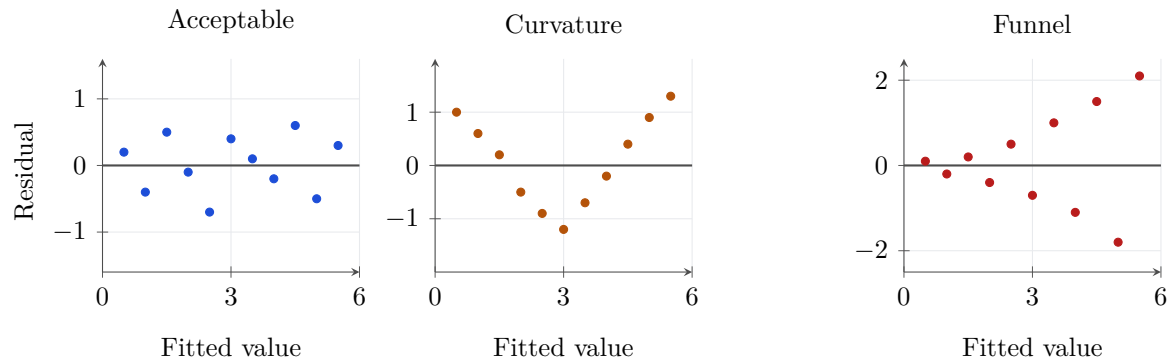
If the model is doing a reasonable job, residuals should usually look like unsystematic noise rather than a visible pattern.

**Residual logic**

The original graph asks whether the relationship looks plausible. The residual plot asks whether the errors left behind by the model look random enough.

5 Residual plots: three common patterns

A residual plot places fitted values or the explanatory variable on the x-axis and residuals on the y-axis. The horizontal zero line represents perfect prediction.



Residual pattern	Possible meaning
Random cloud around zero	The fitted model may be adequate. This is not proof; it is a diagnostic sign.
Curved pattern	A straight-line model may be inappropriate. Consider a nonlinear model or transformation.
Funnel shape	Spread changes with fitted value. Standard errors and p-values from the simple model may be unreliable.

6 Equal spread: homoscedasticity

Many common tests assume that residual spread is roughly similar across the range of fitted values or across groups. This is called **homoscedasticity**. If the spread changes strongly, the data are **heteroscedastic**.

Chemistry example

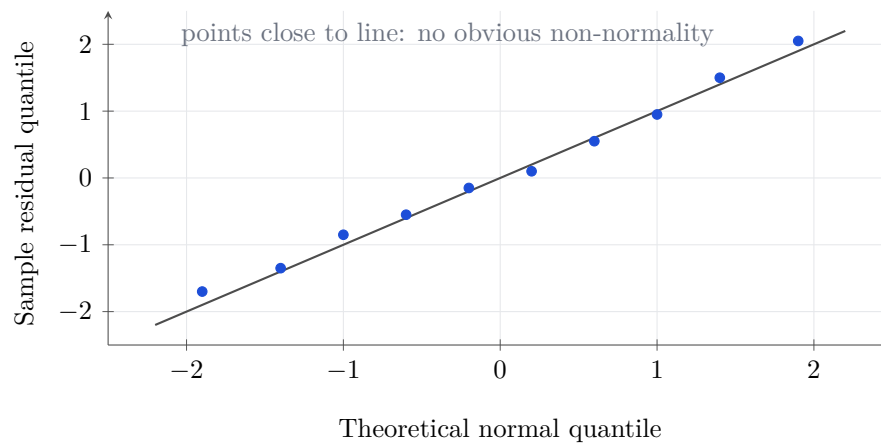
In an absorbance calibration curve, low concentrations may be measured with small absolute error, while high concentrations may show larger scatter because of detector limits, dilution error, or nonlinearity. A simple linear model may still draw a line, but its uncertainty estimates may be misleading.

IB wording

Do not write: “The assumptions are satisfied.” Write a weaker, more honest statement: “The residual plot did not show a strong pattern that would obviously invalidate the model.”

7 Q–Q plots: checking residual shape

A Q–Q plot compares the distribution of residuals with a theoretical distribution, often the normal distribution. If residuals are approximately normal, the points should fall roughly along a straight line.



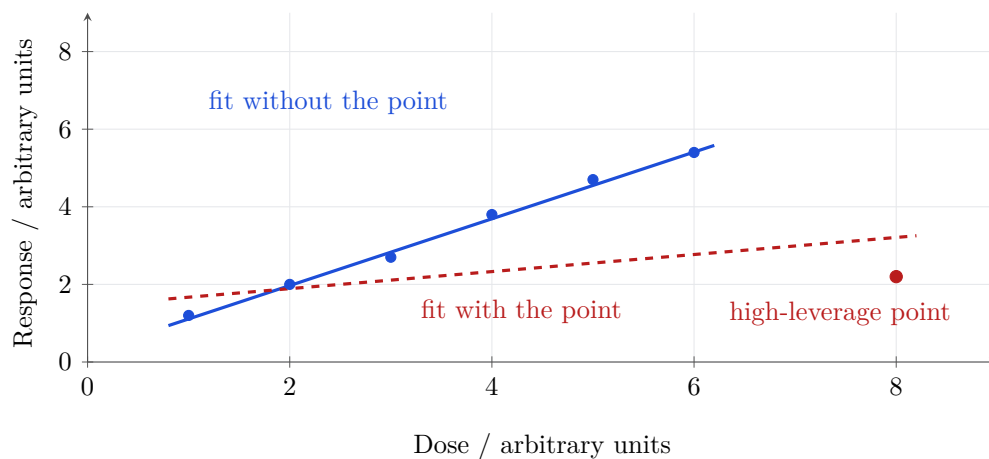
What a Q-Q plot does not do

A Q-Q plot does not prove that residuals are normal. It only helps detect strong departures such as skew, heavy tails, or outliers. With very small samples, it is especially hard to judge.

8 Outliers, leverage, and influence

Students often call every unusual point an outlier. That is too crude. Three ideas are different:

Term	Meaning
Outlier	A point with an unusual response value compared with nearby data.
High leverage	A point with an unusual x-value, far from the main cloud of data.
Influential point	A point that strongly changes the fitted model if removed.



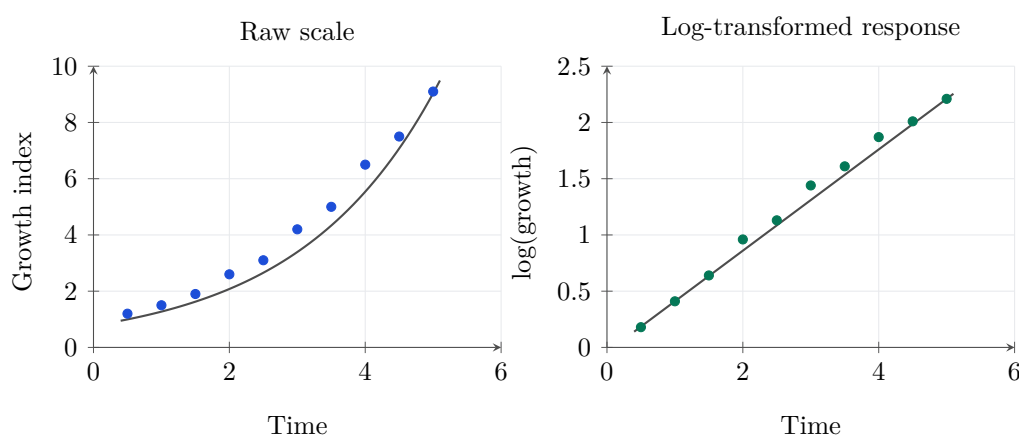
Do not delete data casually

A point should not be removed simply because it is inconvenient. Removal needs a documented reason: contamination, recording error, instrument failure, impossible value, or a pre-defined exclusion rule.

9 Transformations: changing the scale

A transformation changes the scale of the response or explanatory variable. It is not a trick to force significance; it is a way to fit the model to the structure of the data more honestly.

Transformation	Often useful when	Scientific meaning
$\log(y)$	values are positive and right-skewed; error is multiplicative	equal ratios matter more than equal differences
$\sqrt{y}$	count data have variance increasing with mean	stabilizes some count variation
$1/y$ or $1/x$	reciprocal relationships are expected	common in some rate and lens-type relationships

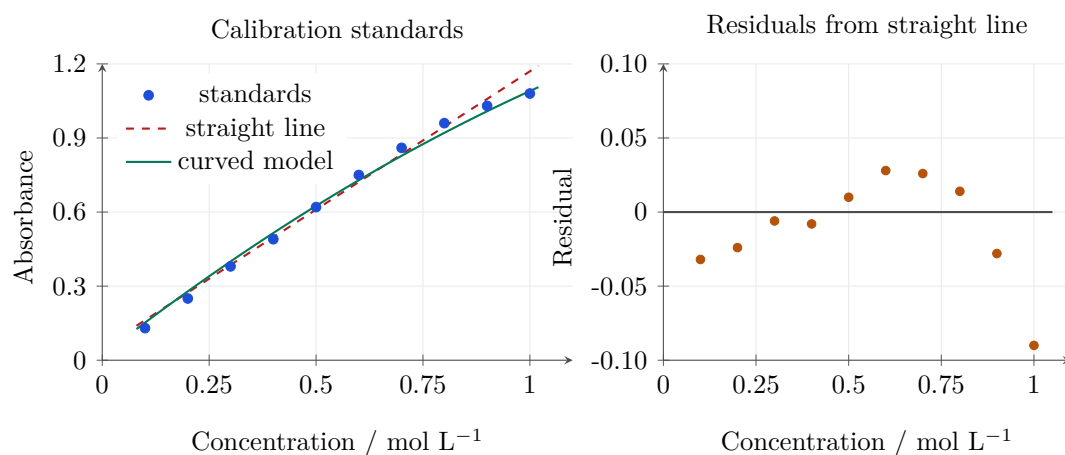


Transformation rule

A transformation should be justified before interpreting the result. After a log transformation, conclusions are naturally about proportional or fold changes, not simple raw-unit differences.

10 Example: linear versus curved calibration

A calibration curve should not be made linear by wishful thinking. First plot the standards. Then check whether the residuals show systematic curvature.

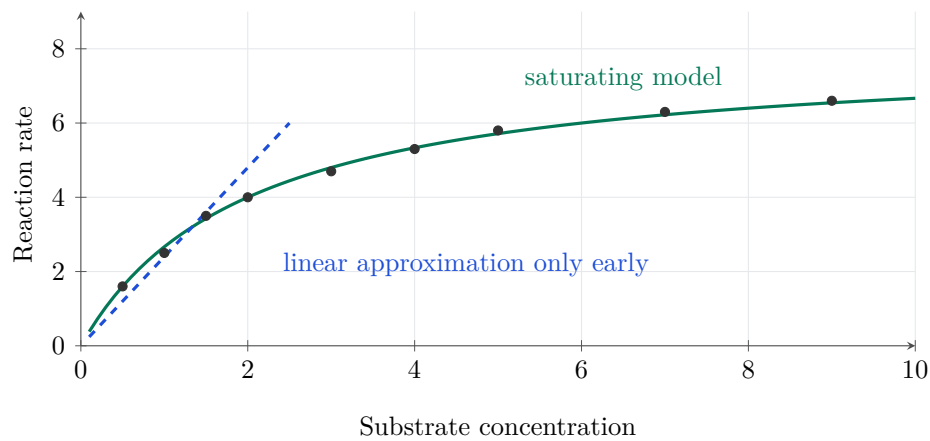


Calibration warning

If the unknown sample lies outside the linear range, do not use the fitted line mechanically. Dilute the sample, narrow the range, or use an appropriate nonlinear model if scientifically justified.

11 Example: enzyme kinetics and model range

A straight line can be reasonable over a narrow range but wrong over a wider range. This is common in enzyme kinetics, growth curves, and absorbance at high concentration.



Range matters

A model can be appropriate within a tested range and inappropriate outside it. This is why extrapolation is dangerous even when the fitted line looks mathematically clean.

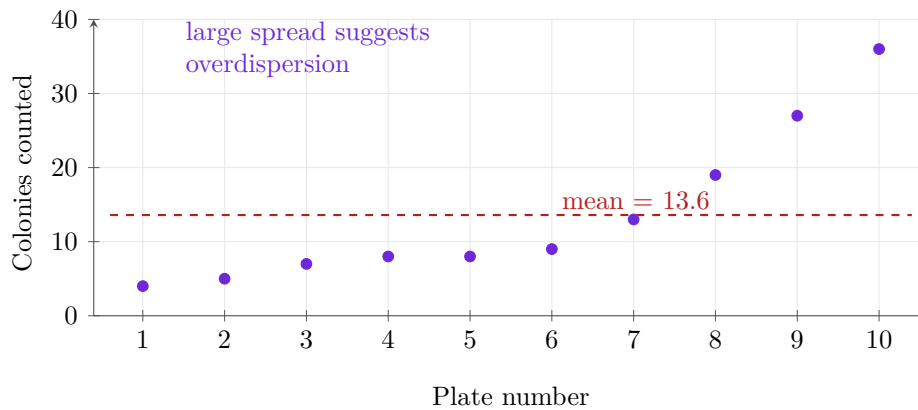
12 Count data and overdispersion

For count data, the distributional assumption matters. A Poisson model has a strong property:

$$\text{mean} \approx \text{variance}.$$

Biological counts often violate this because organisms cluster, plates differ, colonies merge, or samples have unequal exposure.

Plate colony counts	4, 5, 7, 8, 8, 9, 13, 19, 27, 36
Mean	13.6
Variance	109.4
Interpretation	Variance is much larger than the mean, so a simple Poisson model is suspicious.



Biological interpretation

Overdispersion is not just a statistical nuisance. It may reveal real biology: clumping, patchy distribution, uneven growth conditions, or heterogeneous susceptibility.

13 Sensitivity analysis

A sensitivity analysis asks:

Would my conclusion change if I used another plausible analysis?

This is especially useful when assumptions are imperfect or sample size is small.

Concern	Sensitivity check
Unequal variances between groups	Compare classical t-test/ANOVA with Welch version.
Small sample and doubtful normality	Compare parametric test with permutation or non-parametric alternative.
Skewed positive data	Compare raw-scale and log-scale analysis; interpret scale carefully.
Possible influential point	Report analysis with and without the point only if the exclusion has a defensible reason.
Curved relationship	Compare straight-line fit with scientifically plausible nonlinear or transformed model.

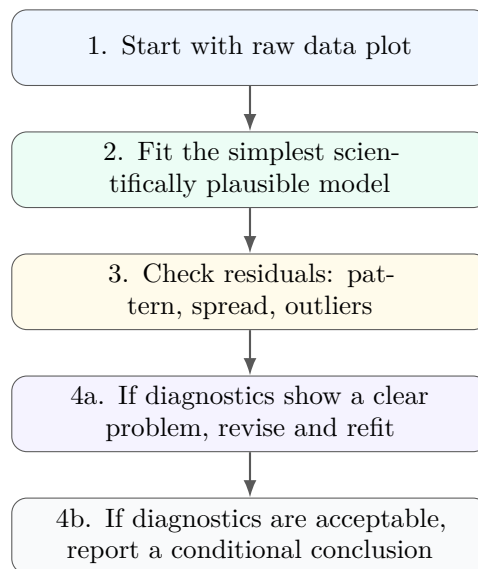
Good reporting phrase

The conclusion was similar under the main analysis and a sensitivity check. This supports the robustness of the interpretation.

Bad reporting phrase

Do not say: "I tried several tests and reported the significant one." That is analysis shopping. If several methods were considered, explain why the final method was chosen.

14 Decision guide



15 Writing model-aware conclusions

Weak conclusion:

The model was significant, so the hypothesis was proven.

Stronger conclusion:

The data showed a positive relationship between concentration and absorbance within the tested range. The residual plot did not show strong curvature in this range, so a linear calibration model was treated as reasonable. The conclusion should not be extrapolated beyond the standards used, and high-concentration samples should be diluted before prediction.

Mini practice

For each situation, name the diagnostic concern and a better next step.

1. A residual plot from a calibration curve forms a U-shape.
2. A Q-Q plot shows two extreme points far from the line.
3. Colony-count variance is much larger than the mean.
4. A single high-dose point changes the regression slope sharply.
5. A log transformation makes the residual spread much more even.

16 Unit 10 summary

Idea	Practical meaning
Model choice	A hypothesis about the data-generating process, not a mere software option.
Residual plot	Main diagnostic for linearity, equal spread, and unusual patterns.
Q–Q plot	Diagnostic for residual shape, especially strong non-normality or tail problems.
Outlier handling	Requires a documented scientific or procedural reason.
Transformation	Changes interpretation; useful only when scientifically justified.
Sensitivity analysis	Checks whether the conclusion survives reasonable alternative analyses.
Model-aware conclusion	States what the data support, within the tested range and under stated assumptions.

Final sentence for students

A model does not make data true. It creates a disciplined way to ask whether a proposed scientific claim is reasonable under uncertainty.